IMPLEMENTATION GUIDE

ServiceNow Import Sets & Transform Maps

Enterprise Spreadsheet Data Migration Architecture & Execution Plan

1. Architecture Overview



PROJECT SCOPE & TARGET OBJECTIVES

This project outlines the operational workflow for ingesting legacy hardware asset records from an external spreadsheet (`Hardware_Inventory_Import.xlsx`) into ServiceNow's core Hardware Asset table (`alm_hardware`), enforcing strict validation and deduplication.

SOURCE DATA SPECIFICATIONS

Asset ID	Device Name	Category	Serial Number	Cost	Purchase Date	Assigned To Email
HW-1001	MacBook Pro 16	Laptop	C02XL012M930	2100	2024-01-15	john.doe@example.com
HW-1002	Dell XPS 15	Laptop	DL-8839210-X	1850	2024-02-10	jane.smith@example.com
HW-1003	ThinkPad X1	Laptop	TP-0091827-Y	1600	2024-03-01	rahul.sharma@example.com

2. Data Source & Staging Setup

STEP 1: CONFIGURE SERVICENOW DATA SOURCE RECORD

Navigate to **System Import Sets > Administration > Data Sources** and create a new record configured as follows:

FIELD PARAMETER

Name

Hardware Inventory Excel Import

FIELD PARAMETER

Import Set Table Label

Hardware Import Staging

FIELD PARAMETER

Type / Format

File / Excel (.xlsx)

FIELD PARAMETER

Header Row / Sheet

Row 1 / Sheet Number 1

STEP 2: FILE ATTACHMENT & RAW DATA INGESTION

1. Attach `Hardware_Inventory_Import.xlsx` to the Data Source form using the paperclip icon.
2. Save the record, then click the **Load All Records** related link.
3. Verify that ServiceNow automatically creates the staging table `u_hardware_import_staging` and generates matching fields.

STAGING VERIFICATION METRICS

Import Set Number	State	Processed Count	Errors
ISET0010042	Loaded	3 Records	0

3. Field Mapping & Transformation Setup

TRANSFORM MAP HEADER PROPERTIES

Navigate to **System Import Sets > Create Transform Map** to establish the structural mapping rule between staging and production tables.

- **Name:** Hardware Staging to Alm Hardware Map
- **Source Table:** Hardware Import Staging [u_hardware_import_staging]
- **Target Table:** Hardware [alm_hardware]
- **Active:** True

FIELD MAPPING MATRIX

Source Staging Field	Target Field	Coalesce Key	Choice Action
u_asset_id	asset_tag	TRUE	N/A
u_device_name	display_name	FALSE	N/A
u_category	model_category	FALSE	Ignore
u_serial_number	serial_number	FALSE	N/A
u_cost	cost	FALSE	N/A
u_purchase_date	purchase_date	FALSE	N/A

COALESCE FUNCTIONALITY EXPLANATION

Setting u_asset_id as **Coalesce = True** causes ServiceNow to search the target table for an existing record matching the Asset Tag. If a match is found, the existing record is updated; if no match exists, a new record is created. This prevents duplicate entries upon repeated runs.

4. Advanced Scripts & Validations

1. USER REFERENCE LOOKUP SCRIPT

Maps the email string in the spreadsheet (`u_assigned_to_email`) to a Sys ID on the User table (`sys_user`).

```
// Field Map Script: Source = u_assigned_to_email, Target = assigned_to
answer = (function transformEntry(source) {
    if (!source.u_assigned_to_email) return "";

    var grUser = new GlideRecord('sys_user');
    if (grUser.get('email', source.u_assigned_to_email.trim().toLowerCase())) {
        return grUser.getValue('sys_id');
    }
    return ""; // Leave blank if user email is not found
})(source);
```

2. ONBEFORE SCRIPT (VALIDATION & FORMATTING)

Runs prior to record insertion to validate mandatory attributes and clean raw string formats.

```
// Transform Script: When = onBefore
(function runTransformScript(source, map, log, target) {
    // Reject processing if primary Asset Key is missing
    if (JSUtil.nil(source.u_asset_id)) {
        ignore = true;
        log.warn("Row skipped: Missing mandatory u_asset_id key.");
    }

    // Standardize Serial Numbers to uppercase without whitespace
    if (!JSUtil.nil(source.u_serial_number)) {
        target.serial_number = source.u_serial_number.toString().trim().toUpperCase();
    }
})(source, map, log, target);
```

5. Execution & Quality Assurance

EXECUTING THE TRANSFORMATION

1. Navigate to **System Import Sets > Run Transform**.
2. Select the active Import Set: `ISET0010042`.
3. Move `Hardware Staging to Alm Hardware Map` into the Selected Maps slushbucket.
4. Click **Transform**.

POST-IMPORT VALIDATION CHECKLIST

- ✓ **Log Verification:** Check Transform History log for state `Complete` with zero warnings or errors.
- ✓ **Coalesce Check:** Re-run the transform job with duplicate data to ensure zero extra rows are created in `alm_hardware`.
- ✓ **Target Table Audit:** Open `alm_hardware` to verify that serial numbers are formatted in uppercase and costs match.
- ✓ **Reference Linking:** Confirm `assigned_to` reference field links directly to the correct `sys_user` record card.

ROLLBACK & RECOVERY PLAN

If malformed data affects target tables during testing:

Navigate to **System Import Sets > Import Sets**, open the specific Import Set record, and click the **Rollback** button to automatically purge all created and updated records.